

DSA - 0405 Fundamental OF Data Science

1. What is Data Science? Explain its Primary Objective

Data Science is the field of extracting meaningful knowledge and insights from data using statistics, machine learning, and computing.
Primary objective: To analyze data and support better decision making through predictions and insights.

2. Why is Data Science important in today's digital world? Mention any four reasons.

- Helps in better decision-making
- Predicts future trends and customer behavior
- Automates business processes

3. List any Four benefits of Data Science in business and society.

- Better business decisions
- Fraud detection
- Improved healthcare services
- Personalized recommendations

4. Mention Four real-world applications of Data Science.

- Healthcare (Disease Prediction)
- Banking (Fraud detection)
- E-commerce (Product recommendation)
- Transportation (Traffic prediction)

5. What are the different facets of data? Briefly explain each

- Structural Data: Stored in rows and columns.
- Semi-Structured Data: Has tags but no fixed format.
- Unstructured Data: No predefined format.

6. Differentiate between Structured, Semi-Structured, and unstructured data with suitable example.

Type	Description	Example
Structured	Fixed Format	Student database
Semi-Structured	Partial Structure	XML, JSON
Unstructured	No Fixed Format	Image, video, 50 text posts

7. What is Big Data? State its key characteristics (5 Vs)

Big Data refers to extremely large and complex datasets that cannot be processed using Traditional methods

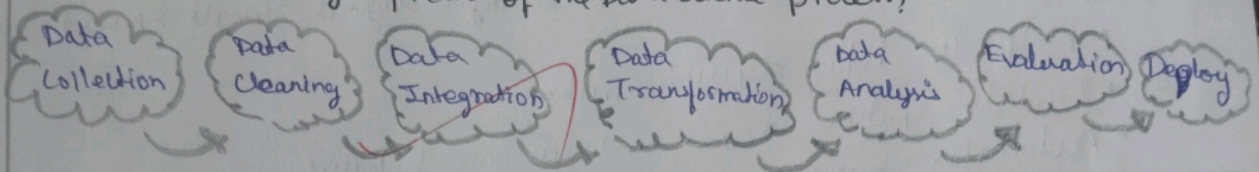
5 Vs : Volume, Velocity, Variety, Veracity, Value

8. Explain The Big Data ecosystem Name any four technologies used in it.

The Big Data ecosystem consists of tools used for collecting, storing, processing and analyzing large datasets.

Technologies : Hadoop, Spark, Hive & HBase

9. What are The Major phase of the Data Science process?



10. What is data Retrieval? Mention Two Common Sources from which data

Data retrieval is the process of obtaining data from various sources
(i) Databases (ii) APIs / websites

11. Why is data collection considered an important step in the Data Science?

- Provides quality data of analysis and Improves prediction accuracy
- Reduces errors and supports reliable decision-making.

12. What is data cleansing? List any four common data-cleaning Techniques.

Data cleansing is the process of correcting or removing inaccurate

Techniques : (i) Remove duplicate (ii) Handle Missing value (iii) correct data (iv) remove outliers

13. What is data Integration? Why is it necessary in Data Science.

Data integration is combining data from multiple sources.

- Provides complete information, Improves analysis, Eliminations a better decision.

14. Explain data Transformation with Suitable example.

Data Transformation is converting data into a suitable format for analysis Ex : Converting text to number, normalizing Numerical value

• converting date formats (dd/mm/yyyy → yyyy-mm-dd)

15. What is data analysis? Mention any four techniques.

Data Analysis is the process of examining data to discover useful information and patterns. Techniques : Descriptive, Diagnostic, Predictive and Prescriptive Analysis